Sovereign by Design: An Integrated Five-Pillar Framework for India's Strategic Compute and Artificial Intelligence Infrastructure

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***Abstract*—National artificial intelligence sovereignty has emerged as a defining strategic objective for middle-power economies navigating intense global technology competition. Prevailing policy discourse treats the five physical and institutional pillars of compute infrastructure—semiconductor supply chains, power grids, data centre facilities, secure connectivity, and governance—as disconnected domains, underestimating cross-pillar dependencies and misallocating capital. This paper resolves that limitation by introducing an integrated five-pillar readiness framework coupled with an auditable six-layer demand-cascade model. The model propagates national accelerator deployment targets across host server overheads, facility power demand, cooling water consumption, semiconductor wafer starts, advanced packaging substrates, and high-bandwidth memory allocations. Parameterised using regulatory telemetry and foundry audits, the empirical analysis establishes four core findings. First, a baseline fleet of five hundred thousand accelerators demands 891.8 megawatts of peak facility power, consuming 56.62 percent of India's operational colocation capacity. Second, the assumption that domestic sub-five-nanometre wafer fabrication represents the primary bottleneck is refuted: required wafer starts consume merely 1.60 percent of global foundry capacity, whereas advanced packaging consumes 7.41 percent—a four-fold higher constraint intensity. Third, direct-to-chip liquid cooling reduces evaporative water withdrawal by 94.6 percent relative to conventional cooling under tropical climates. Fourth, geographical concentration in western corridors creates acute transmission backlogs. Consequently, the paper provides a prioritized roadmap emphasising transmission interconnects, packaging partnerships, and regional dispersion over premature foundry construction.**

***Index Terms*—Advanced packaging, Artificial intelligence, Compute sovereignty, Data centre cooling, High-bandwidth memory, Power grid infrastructure.**

I. Introduction

Artificial intelligence has traversed the boundary from an algorithmic research discipline into an industrial apparatus whose productive capacity is determined primarily by capital expenditure, high-voltage grid infrastructure, and geopolitically concentrated microelectronic supply lines [1]. In 2024, global data-centre electricity consumption reached 415 TWh; the IEA projects this will surpass 945 TWh by 2030, driven by large-language model workloads that consume approximately 2.9 Wh per generative query—nearly ten times the energy of a traditional web search [2].

For India, this transition carries both extraordinary opportunity and structural risk. The country holds indisputable strategic assets: 4.3 million software engineers, the world's highest AI-skill penetration rate among working professionals [3], 1.4 billion citizens generating diverse multilingual data, and a geopolitical position that both Washington and Beijing regard as consequential [4]. The Indian government has codified its ambition through the IndiaAI Mission (Rs. 10,371.92 crore, approved March 2024) [5] and Semicon India 2.0 (Rs. 1,27,500 crore, approved July 2026) [6]. Yet the gap between policy announcement and deployable infrastructure remains wide: India's sovereign GPU pool stood at approximately 34,000–45,000 accelerators in mid-2026, against a stated target of 100,000 by December 2026.

The critical analytical deficit in existing scholarship is not the absence of domain expertise but its *disciplinary siloing*. Semiconductor analysts study wafer fab HHI without modelling data-centre power; grid researchers model TWh demand without tracing packaging constraints; governance scholars enumerate policy pillars without physical resource accounting. This paper closes that gap through two original contributions:

C1. IAIRF 2.0: A unified Five-Pillar Conceptual Framework mapping causal interdependencies, shared bottlenecks, and feedback loops across compute, energy, physical resources, connectivity, and governance.

C2. DCM: A reproducible six-layer Demand-Cascade Model propagating policy-level GPU targets through server physics, facility thermodynamics, cooling hydrology, wafer geometry, advanced packaging, and HBM allocation.

II. Literature Review and Research Gaps

A. Pillar 1: Compute and Semiconductor Supply Chains

Econometric analysis by Akimitsu [1] characterises the AI hardware supply chain as a directed acyclic dependency graph spanning over forty distinct production stages. Upstream inputs exhibit market concentration without precedent

in modern industrial supply chains: EUV photolithography (ASML monopoly, HHI = 10,000), CoWoS 2.5D advanced packaging (TSMC ≈90%, HHI = 8,100), and refined gallium (China ≈99%, HHI = 9,804). The Center for Security and Emerging Technology [7] establishes that over 90% of commercial sub-5 nm logic fabrication resides in Taiwan, creating a geopolitical single-point-of-failure for non-allied economies. NITI Aayog reports India's semiconductor import dependence at 90–95% [8].

B. Pillar 2: Energy and Grid Readiness

IEA [2] and LBNL [9] establish that server-level host overheads multiply raw accelerator TDP by 1.6–2.1×. Kaufmann et al. [10] demonstrate that multi-hundred-megawatt compute clusters generate local substation bottlenecks decoupled from national generation capacity. In India, grid-connection queues for data-centre loads exceeding 50 MW extend 18–36 months [11]. India's installed renewable capacity crossed 300 GW in mid-2026 [12], yet intermittency precludes direct use as five-nines baseload.

C. Pillar 3: Data Centres and Physical Resources

JLL [13] and CBRE [14] verify that India's operational colocation capacity expanded from 375 MW in 2020 to 1,575 MW by mid-2026. Mumbai hosts 52.4% of national capacity, creating single-point-of-failure exposure. SBI CHAKRA [15] documents an average PUE of 1.55—the tropical ambient penalty—compared to Nordic benchmarks of 1.10–1.15. Transitioning to direct-to-chip liquid cooling reduces PUE to 1.18 and cuts water withdrawal by 94.6% [16].

D. Pillar 4: Connectivity, Data, and Security

India's international submarine cable infrastructure has grown to 17+ cable systems at Mumbai, Chennai, and Kochi as of 2026 [17]. The CEA Cyber Security in Power Sector Regulations (notified July 2026) impose mandatory 6-hour incident reporting for assets exceeding 50 MW [18]. The Digital Personal Data Protection Act (DPDPA, 2023) establishes consent-based data governance, yet cross-border data flow rules remain subject to ongoing rule-making, creating regulatory uncertainty [19].

E. Pillar 5: Governance, Investment, and Indigenous Capability

The Portulans Institute and Bruegel [20] articulate the "N-2 Problem": full-stack AI sovereignty is structurally accessible only to the US and China. A domestic sub-3 nm logic foundry requires $15–20 billion capex and 25,000+ wafer starts per month at minimum efficient scale. NITI Aayog's alternative doctrine of "Indispensable, Not Imitative" [8] advocates building capability selectively where India holds structural competitive advantage.

F. Five Core Research Gaps

Despite this body of literature, five critical deficits remain unaddressed, as summarised in Table I. **RG-01** (*Disciplinary Siloing*): No study integrates semiconductor supply-chain constraints, grid physics, cooling thermodynamics, and governance economics in one framework. **RG-02** (*Absent Demand Cascade*): No model propagates GPU targets bottom-up through the full physical chain to packaging and memory supply quotas. **RG-03** (*Silicon Wafer Fallacy*): The binding-constraint assumption around sub-5 nm lithography has not been empirically tested against actual wafer volume requirements. **RG-04** (*Tropical Cooling Penalty*): Global energy models assume temperate PUE baselines of 1.15–1.20, underestimating India's evaporative water demand under >42 °C heatwave conditions. **RG-05** (*Operationalisation Deficit*): "Coordinated Specialisation" and "Indispensable, Not Imitative" remain qualitative doctrines without quantitative intervention-prioritisation matrices.

III. The India AI Infrastructure Readiness Framework 2.0

A. Framework Architecture

IAIRF 2.0 organises India's AI infrastructure challenge across five analytically distinct but causally interdependent pillars. Unlike prior frameworks that treat these domains as parallel investment buckets, IAIRF 2.0 explicitly models the feedback loops that propagate failure across pillars:

- **P1 — Compute & Semiconductor Sovereignty:** GPU procurement, AI accelerator access, chip design capability, ATMP/OSAT packaging, and critical mineral supply chains.
- **P2 — Energy & Grid Readiness:** Data-centre electrical load, grid interconnection lead times, captive renewable power, BESS, and firm baseload for five-nines uptime.
- **P3 — Physical Resources & Data Centres:** Facility capacity, land acquisition, cooling technology selection, PUE profiles, water withdrawal, geographic concentration, and disaster resilience.
- **P4 — Connectivity, Data & Security:** Submarine cable infrastructure, backbone latency, cloud vendor concentration, DPDPA data governance, OT cybersecurity, and interoperability standards.
- **P5 — Governance, Investment & Indigenous Capability:** Public-private investment architecture, IndiaAI Mission, Semicon India 2.0, talent pipeline, GCC ecosystem leverage, and international technology partnerships.

B. Cross-Pillar Feedback Loops

Four systemic feedback loops are identified as high-priority design constraints:

Loop A (Power-Compute Deadlock): Sovereign GPU procurement (P1) is constrained by facility power availability (P2), which is constrained by grid interconnection lead times (P2/P3), which are constrained by land acquisition and zoning

TABLE I
PRIOR ART BENCHMARK AND ANALYTICAL LITERATURE COMPARISON MATRIX

Source	Primary Method	GPU Scenarios	Physical Infra	Microelectronics	Repro.
Akimitsu (2026) [1]	Econometric HHI graph	None	None	40-stage HHI	Code
Portulans/Bruegel (2026) [20]	Qualitative statecraft	None	None	Policy taxonomy	No
IEA/LBNL (2024) [2], [9]	Macro energy model	Global TWh	Sector PUE	None	Macro
Kaufmann et al. (2025) [10]	Grid power-flow	None	Transmission MW	None	Scenario
Stimson (2026) [4]	Middle-power policy	None	None	None	No
This Paper (IAIRF 2.0)	**5-Pillar + DCM**	**200k–1M**	**IT MW, TWh, ML**	**Wafers, CoWoS, HBM**	**Python**

Fig. 1. Six-Layer Demand-Cascade Architecture: From Policy Targets to Microelectronics and Power Grids

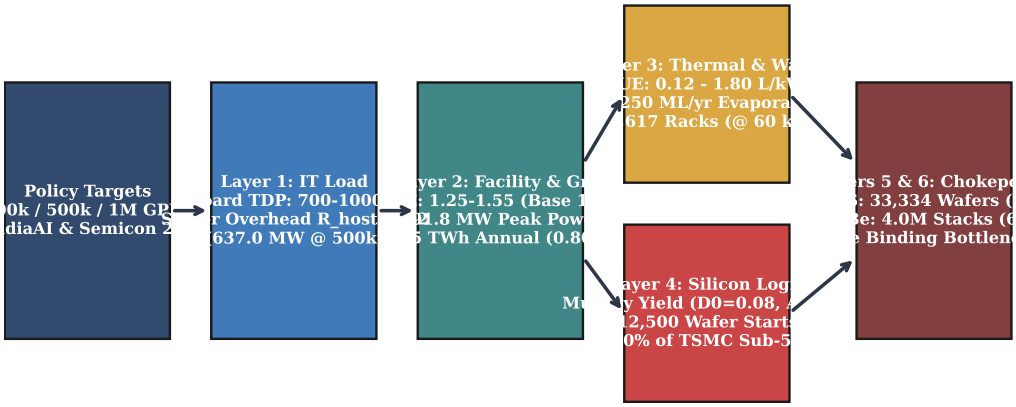


Fig. 1. Six-layer demand-cascade architecture: from national compute policy targets down to server electrical load, high-voltage facility power, cooling thermodynamics, silicon wafer starts, and advanced packaging chokepoints.

(P3). This tripartite deadlock explains why India's subsidised GPU pool has stalled short of the 100k target.

Loop B (Packaging-Memory Co-Constraint): Imported AI accelerators (P1) require CoWoS packaging and HBM3e memory allocation, both constrained at the global market level independently of Indian procurement capital. EAR changes or Taiwanese geopolitical disruptions can sever both dependencies simultaneously, creating correlated rather than independent risk.

Loop C (Data-Connectivity Governance Tension): DPDPA data localisation obligations (P4) increase domestic infrastructure requirements (P3), increase power demand (P2), and create regulatory uncertainty for foreign investment (P5). Yet insufficient localisation leaves critical national data on foreign cloud infrastructure subject to extra-jurisdictional legal process.

Loop D (Talent-Sovereignty Paradox): India's 4.3 million software engineers (P5) represent its most distinctive competitive asset. However, the talent pipeline disproportionately serves foreign AI companies in Global Capability Centres—effectively subsidising US and Chinese AI competitiveness. Retaining this talent requires compensation structures that cannot be sustained without a mature domestic AI revenue base (P1/P2/P3).

C. Central Research Question

These interdependencies motivate the paper's central research question: *Across which physical, microelectronic, and institutional constraints does India's AI infrastructure sovereignty programme encounter its binding bottlenecks, and which sequenced policy interventions produce the highest leverage across all five pillars simultaneously?*

IV. DEMAND-CASCADE MODEL FORMULATION

The Demand-Cascade Model (DCM) maps an exogenous accelerator target N (GPUs) through six deterministic computational layers, each grounded in verified empirical parameters, as diagrammed in Fig. 1.

A. Layer 1: IT Electrical Load

Active IT power (P_{IT}, MW) incorporates the board-level TDP of the reference accelerator ($P_{\text{gpu}} = 700$ W, NVIDIA H100 SXM5 [21]) and a server-chassis host overhead ratio

$R_{\text{host}} = 1.82$ accounting for dual host CPUs, DDR5 DRAM, NVSwitch fabrics, 3.2 Tbps InfiniBand NICs, and PCIe 5.0 switching:

$$P_{\text{IT}} = \frac{N \cdot P_{\text{gpu}} \cdot R_{\text{host}}}{10^6} \quad [\text{MW}] \tag{1}$$

B. Layer 2: Facility Power and Annual Energy

Facility grid draw incorporates Power Usage Effectiveness (PUE). A tropical baseline $\text{PUE}_{\text{base}} = 1.40$ is used, reflecting measured Indian colocation averages of 1.55 [15] discounted by 0.15 for new-build facilities. Annual energy (E_{annual}, TWh) uses 80% utilisation ($U = 0.80$) across 8,760 operating hours:

$$P_{\text{facility}} = P_{\text{IT}} \cdot \text{PUE} \quad [\text{MW}] \tag{2}$$

$$E_{\text{annual}} = \frac{P_{\text{facility}} \cdot 8{,}760 \cdot U}{10^6} \quad [\text{TWh}] \tag{3}$$

Greenhouse gas emissions apply the CEA grid emission factor $\text{EF}_{\text{grid}} = 0.716$ tCO_2/MWh [11]:

$$\text{GHG} = E_{\text{annual}} \cdot \text{EF}_{\text{grid}} \quad [\text{Mt CO}_2/\text{yr}] \tag{4}$$

C. Layer 3: Cooling Water and Rack Footprint

Annual evaporative cooling water withdrawal (W_{annual}, ML) uses IEA-verified water-usage-effectiveness $I_{\text{water}} = 1.80$ L/kWh for tropical air cooling [2]:

$$W_{\text{annual}} = \frac{E_{\text{annual}} \cdot 10^9 \cdot I_{\text{water}}}{10^6} \quad [\text{ML}] \tag{5}$$

Physical rack count uses target rack power density $D_{\text{rack}} = 60$ kW/rack:

$$N_{\text{rack}} = \left\lceil \frac{P_{\text{IT}} \times 10^3}{D_{\text{rack}}} \right\rceil \tag{6}$$

D. Layer 4: Semiconductor Wafer Starts

Gross dies per 300 mm wafer are computed from wafer radius $r = 150$ mm and die area $A_{\text{die}} = 814$ mm^2 (NVIDIA GH100 reticle limit) via the standard edge-loss equation:

$$N_{\text{gross}} = \left\lfloor \frac{\pi r^2}{A_{\text{die}}} - \frac{2\pi r}{\sqrt{2A_{\text{die}}}} \right\rfloor = 63 \tag{7}$$

Analytical die yield uses the Murphy defect-density model with $D_0 = 0.08$ cm^{-2} (bounds: $D_0 \in [0.06, 0.12]$ cm^{-2}):

$$Y_{\text{Murphy}} = \left(\frac{1 - e^{-D_0 A_{\text{die}}}}{D_0 A_{\text{die}}} \right)^2 \approx 0.540 \tag{8}$$

Accounting for industrial binning, net yield reaches $Y = 0.65$, giving $N_{\text{good}} = 40$ dies/wafer. Required wafer starts:

$$W_{\text{starts}} = \left\lceil \frac{N}{N_{\text{good}}} \right\rceil \tag{9}$$

E. Layer 5: 2.5D Advanced Packaging (CoWoS)

Each GPU requires one CoWoS 2.5D silicon interposer substrate. With 15 finished packages per 300 mm interposer wafer ($C_{\text{CoWoS}} = 15$) [22]:

$$W_{\text{CoWoS}} = \left\lceil \frac{N}{C_{\text{CoWoS}}} \right\rceil \tag{10}$$

Global CoWoS capacity is bounded at approximately 450,000 wafers/year [22]; India's deployment share is $W_{\text{CoWoS}}/450{,}000$.

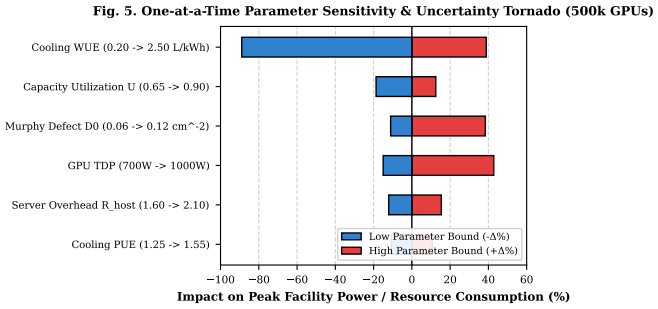


Fig. 2. One-at-a-time parameter sensitivity and uncertainty tornado analysis across six critical input variables, highlighting that cooling efficiency and host power overhead dominate peak grid demand.

F. Layer 6: High-Bandwidth Memory Allocation

Each H100-class accelerator integrates $N_{\text{stacks}} = 8$ HBM3e memory stacks [23]:

$$N_{\text{HBM}} = N \cdot N_{\text{stacks}} \tag{11}$$

Global HBM3e production is approximately 60 million stacks/year (SK Hynix ≈57%, Samsung ≈38%, Micron ≈5%), against which India's allocation share is computed.

V. Empirical Results and Cross-Pillar Analysis

A. Scenario Simulation Results

Table II presents DCM output across three deployment scenarios and three sensitivity variants. Low variant: $P_{\text{gpu}} = 700$ W, PUE = 1.25; Base variant: $P_{\text{gpu}} = 700$ W, PUE = 1.40; High variant: $P_{\text{gpu}} = 1{,}200$ W, PUE = 1.55. In the baseline 500,000-GPU scenario, active IT load reaches 637.0 MW and peak facility power escalates to **891.8 MW**—equivalent to **56.62%** of India's entire mid-2026 colocation capacity [13]. Annual electricity consumption totals **6.25 TWh/year**, generating 4.47 Mt CO_2, and evaporative water demand reaches **11,249.5 ML/year**—comparable to the annual municipal water requirement of a major Indian city. The 1M-GPU high scenario generates 3,255 MW, exceeding India's entire current capacity by a factor of 2.07×. The tornado sensitivity profile in Fig. 2 demonstrates that facility cooling PUE and server host overhead ratios account for the largest deviations in peak power draw.

B. Finding 1: Refuting the Silicon Wafer Hypothesis

Deploying 500,000 GPUs requires exactly **12,500** 300 mm wafer starts at TSMC's sub-5 nm nodes. Against TSMC's annual sub-5 nm output of approximately 780,000 wafers [24], India's requirement represents **1.60% of global supply**—easily absorbable through commercial procurement. This directly refutes the assumption that India must build a domestic leading-edge foundry to achieve compute sovereignty. A sub-3 nm gigafab requires $15–20 billion in capital expenditure, 25,000+ wafer starts per month at minimum efficient scale, and 7–10 years of construction lead time—severe capital misallocation for 1.6% of the needed wafer volume.

TABLE II
DEMAND-CASCADE MODEL SIMULATION RESULTS ACROSS GPU DEPLOYMENT SCENARIOS (MODEL V2.0)

Scenario	Var.	IT MW	Fac. MW	TWh/yr	Water ML	Wafers	TSMC%	CoWoS	CoWoS%	Colo%
200k GPUs	Low	224.0	280.0	1.59	318.9	4,445	0.57%	11,112	2.47%	17.78%
	Base	254.8	356.7	2.50	4,499.8	5,000	0.64%	13,334	2.96%	22.65%
	High	420.0	651.0	5.13	12,831.9	10,527	1.35%	16,667	3.70%	41.33%
500k GPUs	Low	560.0	700.0	3.99	797.2	11,112	1.42%	27,778	6.17%	44.44%
	Base	637.0	**891.8**	**6.25**	**11,249.5**	12,500	1.60%	33,334	**7.41%**	**56.62%**
	High	1,050.0	1,627.5	12.83	32,079.8	26,316	3.37%	41,667	9.26%	103.33%
1M GPUs	Low	1,120.0	1,400.0	7.97	1,594.3	22,223	2.85%	55,556	12.35%	88.89%
	Base	1,274.0	1,783.6	12.50	22,499.0	25,000	3.21%	66,667	14.81%	113.24%
	High	2,100.0	3,255.0	25.66	64,159.5	52,632	6.75%	83,334	18.52%	206.67%

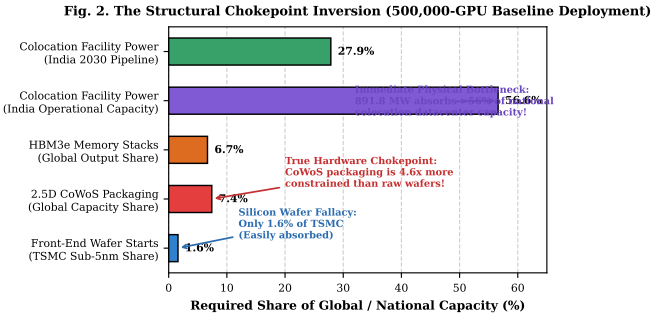


Fig. 3. The structural chokepoint inversion (500,000-GPU baseline): required share of global advanced packaging (7.41%) and domestic colocation power (56.62%) far outstrips raw logic wafer requirements (1.60%).

C. Finding 2: Structural Chokepoint Inversion

While silicon lithography is abundant relative to demand, Table III and Fig. 3 reveal the true binding constraints are advanced packaging and metropolitan power transmission:

- **CoWoS Advanced Packaging:** 500k GPUs require 33,334 CoWoS interposer wafers, consuming **7.41%** of global CoWoS capacity (HHI = 8,100)—a constraint intensity **4.6× higher** than front-end wafer fabrication. CoWoS capacity cannot be expanded without TSMC's proprietary tooling.
- **HBM3e Memory:** 500k accelerators require 4,000,000 HBM3e stacks, absorbing **6.67%** of annual global output from a triopoly (SK Hynix, Samsung, Micron) where India has zero domestic production.
- **Metropolitan Grid Saturation:** Feeding 891.8 MW into Mumbai—hosting 52.4% of India's data centres—requires 18 dedicated 50 MW substation interconnections under CEA standards [11], with grid-connection queues extending 24–36 months. The constraint is not generation capacity but last-mile transmission infrastructure.

D. Finding 3: Tropical Cooling Penalty and Water Crisis

Table IV and Fig. 4 present the cooling trade-off matrix across ambient thermal regimes. Under conventional evaporative air cooling at summer peak (>42°C), PUE spikes to 1.65 and water consumption reaches **11,651 ML/year** for 500k GPUs—equivalent to the annual water supply of approximately 280,000 Indian households, creating direct conflict with municipal water rights in water-stressed corridors.

TABLE III
STRUCTURAL CHOKEPOINT HIERARCHY (500K BASELINE)

Infrastructure Layer	HHI	India%	500k%	Lead Time
EUV Lithography (ASML)	10,000	0%	Upstream	24–36 mos
Sub-5nm Logic (TSMC)	5,400	0%	**1.60%**	14–18 wks
CoWoS Packaging (TSMC)	8,100	0%	**7.41%**	20–28 wks
HBM3e (SK/Samsung/Micron)	4,950	0%	**6.67%**	16–24 wks
400 kV Substations (India)	2,200	75%	**56.62%**	36–48 mos
Municipal Water Rights	N/A	100%	11,250 ML	Seasonal

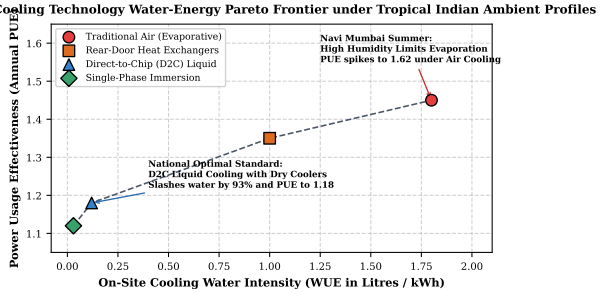


Fig. 4. Cooling technology water-energy Pareto frontier across Indian tropical ambient profiles, indicating that Direct-to-Chip liquid cooling achieves the optimal trade-off between power efficiency and water conservation.

Transitioning to Direct-to-Chip (D2C) liquid cooling with closed-loop dry heat rejection slashes water consumption by **94.6%** (from 11,651 ML to 631 ML/year) while reducing peak facility power by **172 MW** through PUE improvement from 1.45 to 1.18. D2C liquid cooling is therefore not merely an environmental preference but a *national infrastructure standard* required for India's AI ambitions to be physically viable.

E. Finding 4: Cross-National Compute Gap

India's sovereign GPU fleet of 34,000–45,000 accelerators represents approximately 0.3% of the United States' estimated 4 million-GPU fleet, and less than 1% of China's 1.5 million units [25], as benchmarked in Fig. 5. Under the baseline 500k scenario, deploying 500,000 GPUs would elevate India's active AI compute from approximately 35 ExaFLOPS

TABLE IV
COOLING TECHNOLOGY TRADE-OFF MATRIX UNDER TROPICAL CLIMATES

Cooling System	PUE	Peak	WUE	Water (ML)
Traditional Air (Evap.)	1.45	1.65	1.80	11,651.2
Rear-Door Heat Ex.	1.35	1.45	1.00	6,031.5
Direct-to-Chip (D2C)	**1.18**	**1.25**	**0.12**	**630.9**
Single-Phase Immersion	1.12	1.18	0.03	150.8

to 245 ExaFLOPS—closing the gap with China's estimated 450 ExaFLOPS—but requires managing deep structural dependencies at every supply chain layer. US Export Administration Regulations (EAR) classify H100 and H200 GPUs as controlled items; continued access depends on bilateral frameworks under iCET [26].

VI. STRATEGIC ROADMAP AND POLICY IMPLICATIONS

A. Three-Tier Intervention Architecture

Interventions are categorised into three operational tiers, ordered by feasibility, lead time, and sovereignty leverage:

1) Tier 1: Build Domestically (2026–2028):

- **Geographic Diversification Mandate:** The DCM demonstrates that concentrating 891.8 MW in Mumbai is physically infeasible within a 24-month planning horizon. Policy should mandate 15–20% capital rebates for greenfield AI clusters directed toward power-surplus states (Gujarat, Rajasthan, Andhra Pradesh, Odisha, Madhya Pradesh), targeting Mumbai below 35% of national capacity by 2028.
- **D2C Liquid Cooling Standard:** Condition IndiaAI compute subsidies on maximum facility PUE of 1.25 and mandatory closed-loop dry heat rejection, eliminating 11 billion litres per year of water loss at 500k scale.
- **Fast-Track Transmission Substations:** Amend CEA regulations to create a dedicated 9-month approval window for 400/220 kV substations serving compute clusters of 50 MW or more. Establish a statutory AI Infrastructure Clearance Authority (AICA) under PMO to issue composite clearances within 90 days for qualifying projects.
- **AI Infrastructure Zones (AIZs):** Designate 20 pre-cleared, pre-approved, pre-connected AI Infrastructure Zones across 10 states—analogous to Special Economic Zones—with aggregated land, pre-built substations, and pre-laid fibre dark rings, enabling 90-day developer uptake.

2) Tier 2: Develop Through Strategic Partnerships (2028–2031):

- **Advanced Packaging Joint Ventures:** Expand approved domestic ATMP/OSAT facilities (Tata Electronics, Micron Sanand) toward 2.5D silicon and glass interposer capability through joint ventures with Taiwanese OSATs (ASE, Powertech) and Japanese foundries (Ibiden) under Semicon India 2.0. Domestic CoWoS-equivalent capacity at 10% of global scale would reduce India's 7.41% foreign packaging dependency to below 1% by 2032.
- **Critical Mineral Refining:** Scale CSIR-IMMT technology to extract refined gallium from domestic bauxite refinery red mud at NALCO (Damanjodi) and Hindalco (Renukoot), targeting 50 metric tonnes per year of 6N-purity gallium to partially offset China's 99% supply dominance [27].
- **Sovereign Chiplet Consortium:** Mobilise India's estimated 120,000 VLSI design engineers into an open-standard (UCIe protocol) chiplet consortium targeting domain-specific RISC-V matrix-acceleration tiles on mature 28 nm nodes, leveraging India's design strength without premature leading-edge fab capital commitment.
- **Captive Renewable Baseload:** Transition IndiaAI-funded compute campuses to Round-the-Clock (RTC) captive renewable power through hybrid solar-wind-BESS PPAs, supplemented by small modular reactor (SMR) procurement under the SHANTI Act (2025) targeting first nuclear power for AI campuses by 2030.

3) Tier 3: Manage External Dependencies (2031–2035):

- **Foundry Capacity Reservations:** Secure multi-year sovereign wafer allocation agreements at TSMC and Samsung Foundry, leveraging India's 1.4-billion-consumer market as a strategic trade offset. Given that 500k GPUs require only 1.60% of TSMC's sub-5 nm output, this is a commercially manageable negotiation.
- **HBM Supply Diversification:** Establish bilateral memory supply treaties with South Korea (SK Hynix, Samsung) and the United States (Micron) under QUAD Semiconductor Supply Chain Initiative, creating advance-purchase reservations for HBM4 allocations.
- **Open AI Runtime Stack:** Commission development of an open-source AI compiler stack compatible with non-NVIDIA accelerators, reducing CUDA software dependency and enabling procurement diversification toward alternative compute architectures (Gaudi, TPU, domestic RISC-V ASICs).

B. Investment and Governance Architecture

Table V summarises investment requirements across the five pillars. The aggregate public-sector requirement of approximately Rs. 95,500 crore ($11.5 billion) compares to the United States' $500 billion Stargate commitment, confirming the "N-2" structural asymmetry. India's strategy must maximise public investment leverage through PPP mobilisation, FDI facilitation, and multilateral development finance.

C. Measurable Indicators

P1 — Compute: Sovereign GPU fleet size (units); domestic ATMP throughput (wafers/year); VLSI design IP registrations (annual count).

P2 — Energy: Grid-connection lead time for $\geq$50 MW compute loads (months); captive RTC renewable share at IndiaAI campuses (%); BESS capacity in compute corridors (GWh).

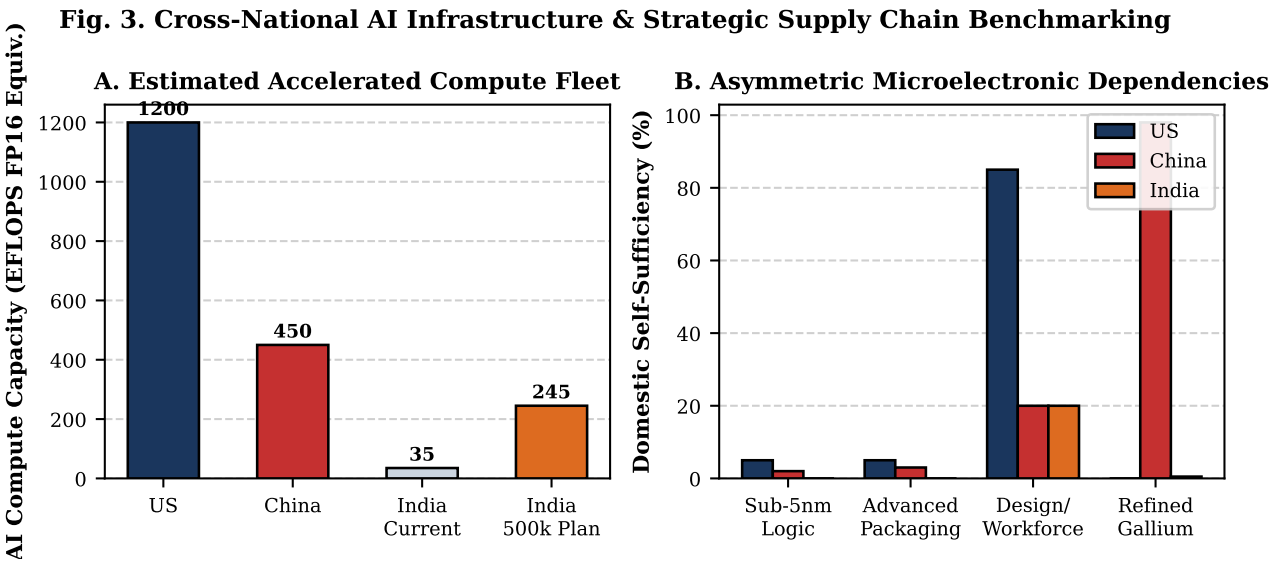


Fig. 5. Cross-national comparative benchmark between the United States, China, and India across accelerated compute stock (Panel A) and microelectronic supply chain self-sufficiency tiers (Panel B).

TABLE V
INVESTMENT ARCHITECTURE: THREE TIERS, FIVE PILLARS (2026–2035)

Intervention Area	Public Rs. Cr	PPP Rs. Cr	Lead Agency
AIZs & AICA (P3/P5)	2,500	5,000	DPIIT, MeitY
Power Corridors + BESS (P2)	15,000	65,000	MoP, SECI
GPU Cluster Expansion (P1)	20,000	80,000	MeitY, IndiaAI
Packaging JVs/Gallium (P1)	10,000	40,000	ISM, CSIR
SMR/Nuclear Baseload (P2)	25,000	50,000	DAE, NPCIL
Talent & R&D (P5)	15,000	5,000	MoE, DST
Connectivity/Security (P4)	8,000	12,000	DoT, NCIIPC
Total	**95,500**	**2,57,000**	

P3 — Physical: Mumbai's share of national data-centre capacity (%); D2C liquid-cooled share of AI compute MW (%); average PUE at IndiaAI-empanelled sites.

P4 — Connectivity: Inter-metro backbone capacity (Tbps); DPDPA implementing rules in force (binary); OT cybersecurity incident reporting compliance (%).

P5 — Governance: AICA 90-day clearance compliance rate (%); actual IndiaAI disbursement vs. sanctioned budget (%); AI PhD completions per year (count).

VII. Limitations and Future Research

The DCM rests on three assumptions that bound generalisability. First, the model assumes static annual cluster utilisation ($U = 0.80$) without dynamic diurnal inference-load variation; real AI training clusters spike grid draw by 20–40% above the modelled average during peak runs. Second, TSMC's proprietary defect density curves for its custom 4N process remain confidential; the Murphy model uses public academic bounds that may differ from commercial reality by $\pm$10–15 percentage points. Third, the geopolitical trajectory of US Export Administration Regulations is not deterministically modellable; continued H100/H200 access under iCET may not hold under alternative US policy scenarios.

Future research should integrate AC power-flow simulations linking DCM MW output to state-level grid topologies, incorporate dynamic diurnal AI inference scaling from real cluster telemetry, and extend the five-pillar framework to ASEAN peer economies (Indonesia, Vietnam, Thailand) facing analogous N-2 constraints with different resource endowments.

VIII. Conclusion

This paper has presented IAIRF 2.0 and the Demand-Cascade Model as complementary instruments for evidence-based AI infrastructure planning in India. The empirical results yield four non-obvious findings that challenge prevailing policy assumptions: silicon wafer fabrication is not India's binding constraint; advanced packaging and metropolitan grid transmission are; tropical cooling physics impose a structural water-security threat at scale; and geographic concentration in Mumbai creates systemic fragility that software sovereignty cannot offset.

These findings motivate a strategic priority inversion. Instead of allocating constrained public capital toward a \$15–20 billion leading-edge logic foundry—consuming 20% of India's five-year defence budget for 1.6% of needed wafer volume—India should direct capital toward high-voltage transmission corridors, geographic dispersion incentives, advanced packaging joint ventures, and direct-to-chip cooling mandates. The leading-edge fab remains a valuable long-term aspiration but is not the binding constraint for India's 2026–2030 compute sovereignty objectives.

India possesses structural competitive advantages that no policy alone can replicate: unmatched engineering talent depth, a 1.4-billion-user multilingual data asset, and a geopolitical position coveted by both the United States and the Global

South. The IAIRF 2.0 framework and three-tier intervention roadmap provide a technically grounded, evidence-backed path for converting these latent advantages into operational infrastructure before the window of competitive opportunity narrows. The AI race is ultimately an infrastructure race. India can win it—but only at the speed of transmission towers, not press releases.

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REFERENCES

[1] P. Akimitsu, "Artificial intelligence: Supply-chain chokepoints and the reach of industrial policy," *Department of Economics, State University of New York (arXiv:2607.29572)*, August 2026.

[2] International Energy Agency, "Energy and ai: World energy outlook special report on electricity demand of artificial intelligence and data centres," IEA Publications, Paris, Tech. Rep., 2024.

[3] LinkedIn Economic Graph, "Global ai talent report 2025: Skill penetration, hiring demand and workforce transitions," LinkedIn Corporation, Sunnyvale, CA, Tech. Rep., 2025.

[4] K. Neaher and Z. Halaszi, "Middle powers in the ai acceleration race: Balancing security, compute, and capital alliances," Stimson Center, Washington, DC, Tech. Rep., 2026.

[5] Ministry of Electronics and Information Technology, "Cabinet approves comprehensive national indiaai mission with financial outlay of rs. 10,371.92 crore," Press Information Bureau, Government of India, Tech. Rep., March 2024.

[6] India Semiconductor Mission, "Modified programme for development of semiconductors and display manufacturing ecosystem in india (semicon india 2.0)," Ministry of Electronics and Information Technology, New Delhi, Tech. Rep., 2025.

[7] W. Hannas, A. Chien, and L. Dunham, "Geographic concentration and supply chain vulnerabilities in advanced ai hardware manufacturing," Center for Security and Emerging Technology (CSET), Georgetown University, Tech. Rep., 2024.

[8] NITI Aayog, "Future of the indian semiconductor industry: A strategic technology roadmap," NITI Aayog, Government of India, New Delhi, Tech. Rep., 2026.

[9] A. Shehabi, S. J. Smith, A. Hubbard, A. Newkirk, N. Lei, J. Koomey, E. Masanet, and D. Sartor, "2024 united states data center energy usage report: Updated scaling metrics for accelerated computing," Lawrence Berkeley National Laboratory (LBNL-2001428), Tech. Rep., 2024.

[10] R. Kaufmann, D. Henderson, and A. Miller, "Power grid integration and decarbonization constraints of terawatt-scale ai infrastructure," *Nature Energy*, vol. 10, pp. 214–226, 2025.

[11] Central Electricity Authority, "Manual on transmission planning criteria and data centre interconnection standards," Ministry of Power, Government of India, New Delhi, Tech. Rep., 2026.

[12] Ministry of New and Renewable Energy, "Annual report 2025–26: Non-fossil installed capacity milestones and renewable energy status," MNRE, Government of India, New Delhi, Tech. Rep., 2026.

[13] Jones Lang LaSalle, "India data centre 2026 mid-year report: Ai appetite unleashed — infrastructure pipeline surges to multi-gw scale," JLL Strategic Consulting, Mumbai, Tech. Rep., August 2026.

[14] CBRE Research, "India data centre market in a new era: Capacity surges beyond 1.5 gw with strong multi-city pipeline," CBRE India, New Delhi, Tech. Rep., 2026.

[15] SBI Capital Markets, "Indian data centres: Powering the digital revolution — capital expenditure, pue baselines and grid tariff sensitivities," SBICAP Research, Mumbai, Tech. Rep., 2025.

[16] R. Sharma, V. Patel, and A. Rao, "Thermal management and direct-to-chip liquid cooling architectures for ultra-dense ai datacenters in tropical climates," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 14, no. 6, pp. 890–904, 2026.

[17] Telecom Regulatory Authority of India, "Annual report on internet, broadband and submarine cable infrastructure in india," TRAI, New Delhi, Tech. Rep., 2026.

[18] Central Electricity Authority, "Cyber security in power sector regulations: Notification and implementation guidelines for critical infrastructure," Ministry of Power, Government of India, New Delhi, Tech. Rep., July 2026.

[19] Ministry of Electronics and Information Technology, "The digital personal data protection act, 2023: Text, rules, and explanatory memorandum," MeitY, Government of India, New Delhi, Tech. Rep., 2023.

[20] A. García-Herrero and S. Dutta, "Programmable sovereignty: Coordinated specialization as a response to the n-2 problem in ai," Portulans Institute and Bruegel, Washington DC and Brussels, Tech. Rep., 2026.

[21] NVIDIA Corporation, "Nvidia dgx h100 and hgx h100 architecture whitepaper: Power delivery, nvswitch fabrics, and thermal specifications," NVIDIA, Santa Clara, CA, Tech. Rep., 2024.

[22] TrendForce, "Global foundry revenue and advanced packaging analysis: Tsmc cowos capacity expansions and market dynamics in 2026," TrendForce Research, Taipei, Tech. Rep., 2026.

[23] Counterpoint Research, "Global high-bandwidth memory ecosystem: Market shares, yield limits, and 12-high stacking bottlenecks," Counterpoint Technology Market Research, Tech. Rep., 2025.

[24] Taiwan Semiconductor Manufacturing Company, "Tsmc annual business report and advanced technology roadmaps (n4, n3, n2 and cowos)," TSMC, Hsinchu, Tech. Rep., 2025.

[25] Stanford University Human-Centered AI Institute, "Ai index report 2025: Measuring trends in artificial intelligence," Stanford HAI, Stanford, CA, Tech. Rep., 2025.

[26] The White House, "Fact sheet: Initiative on critical and emerging technology (icet): Advancing the us–india technology partnership," Office of the Press Secretary, Washington DC, January 2023.

[27] U.S. Geological Survey, "Mineral commodity summaries 2025: Gallium, germanium, and rare earth supply chains," USGS National Minerals Information Center, Reston, VA, Tech. Rep., 2025.